Model diagnostics

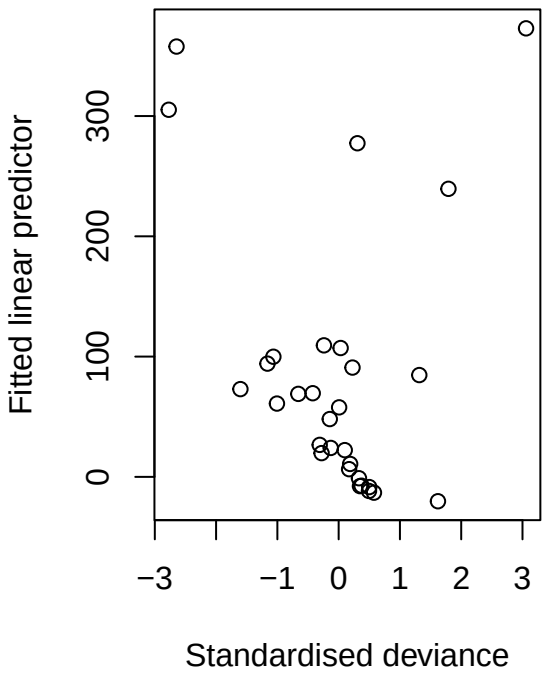
Solutions:

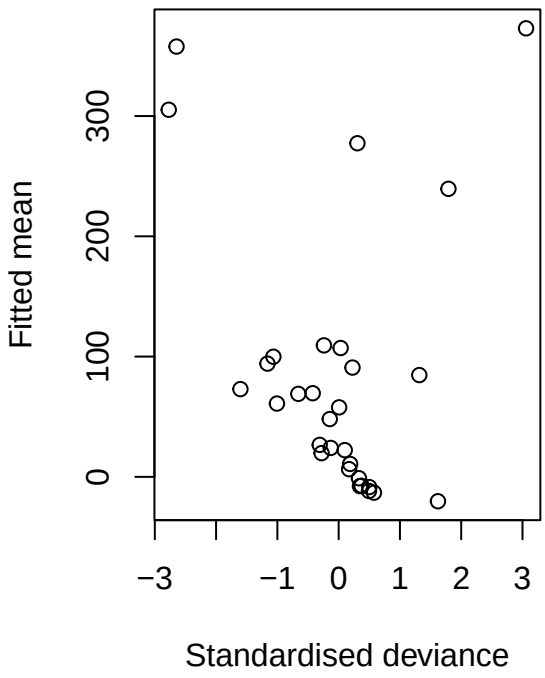
```
checking<-function(ymatrix,xmatrix,famil){
    model1<-glm(ymatrix~.,data=data.frame(ymatrix,xmatrix),family=famil)
    print(summary(model1))
    mu<-predict(model1,type="response")
    eta<-predict(model1,type="link")
    par(mfrow=c(1,2))
    # See Remarks 1-2
    plot(rstandard(model1),mu,
    xlab="Standardised deviance",ylab="Fitted linear predictor")
    plot(rstandard(model1),eta,
    xlab="Standardised deviance",ylab="Fitted mean")
    # See Remark 3
    if (famil=="poisson"){
        z<-log(mu)+(y-mu)/mu
    }
    if (famil=="gaussian"){
        z<-y
    }
    if (famil=="binomial"){
        z<-log(mu/(1-mu))+(y-mu)/(mu*(1-mu))
    }
    if (famil=="Gamma"){
        z<-1/mu-(y-mu)/(mu^2)
    }
    par(mfrow=c(3,dim(xmatrix)[2]/3))
    # See Remark 4
    for (i in 1:dim(xmatrix)[2]){
        temp4<-z-eta+coef(model1)[i]*x[,i]
        plot(temp4,x[,i],ylab=paste("covariate",i),xlab="ei")
        temp<-lm(x[,i]~temp4)
        delta<-(max(temp4)-min(temp4))/10
        x2<-seq(min(temp4)-delta,max(temp4)+delta,by=((max(temp4)-min(temp4))/1000))
        lines(x2,temp$coefficients[1]+temp$coefficients[2]*x2)
    }
    par(mfrow=c(1,1))
    cooks.distance(model1)
    # See Remark 5
    plot(cooks.distance(model1),
    xlab="ith observations",ylab="cook's distance")
    temp<-xmatrix%*%solve(t(xmatrix)%*%xmatrix)%*%t(xmatrix)
    temp<-as.matrix(temp)
    set1<-matrix(0,nrow=1,ncol=nrow(temp))
    for (i in 1:nrow(temp)){
        set1[i]<-temp[i,i]
    }
```

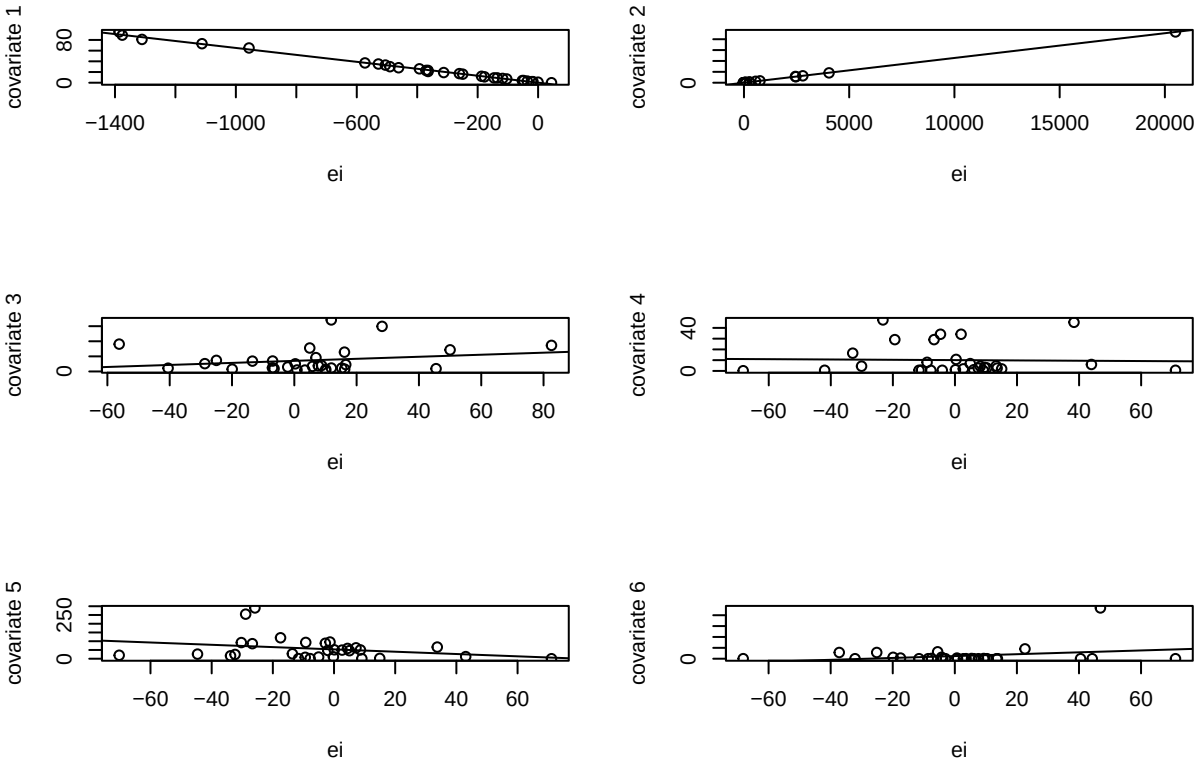
```
    as.array(set1)
    # See Remark 6
    plot(1:length(as.list(set1)),as.list(set1),
    xlab="ith observations",
    ylab="leverage")
}

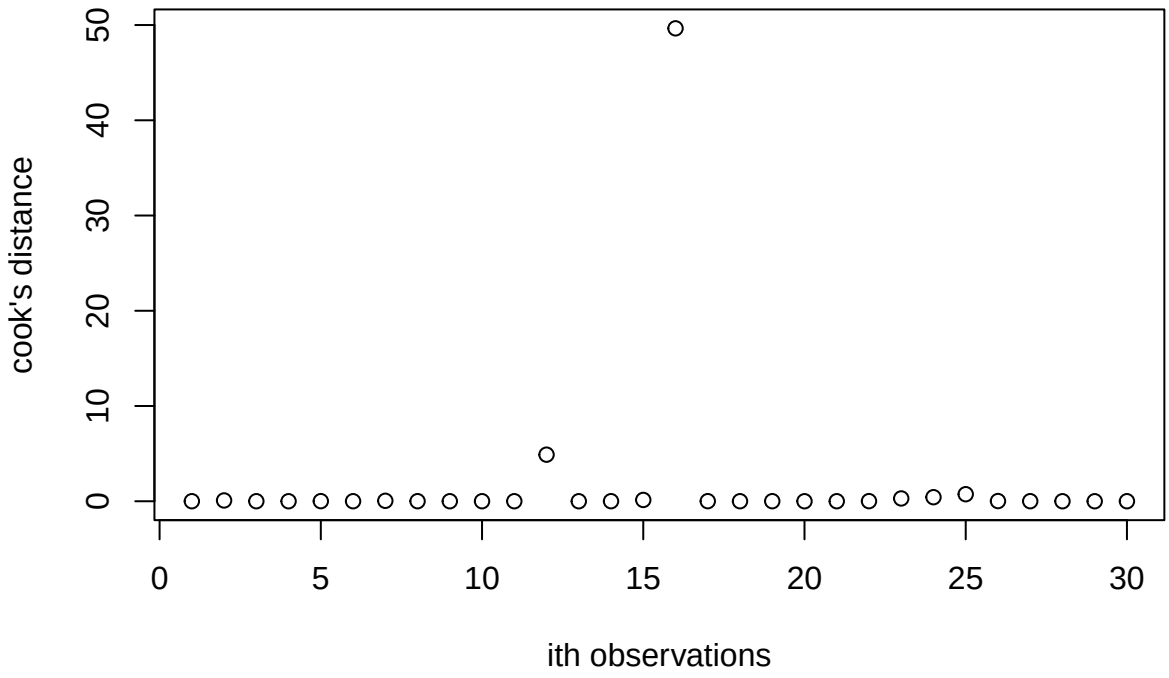
library(faraway)
data(gala)
x<-as.matrix(gala[,2:dim(gala)[2]])
y<-as.matrix(gala[,1])
checking(y,x,"gaussian")
```

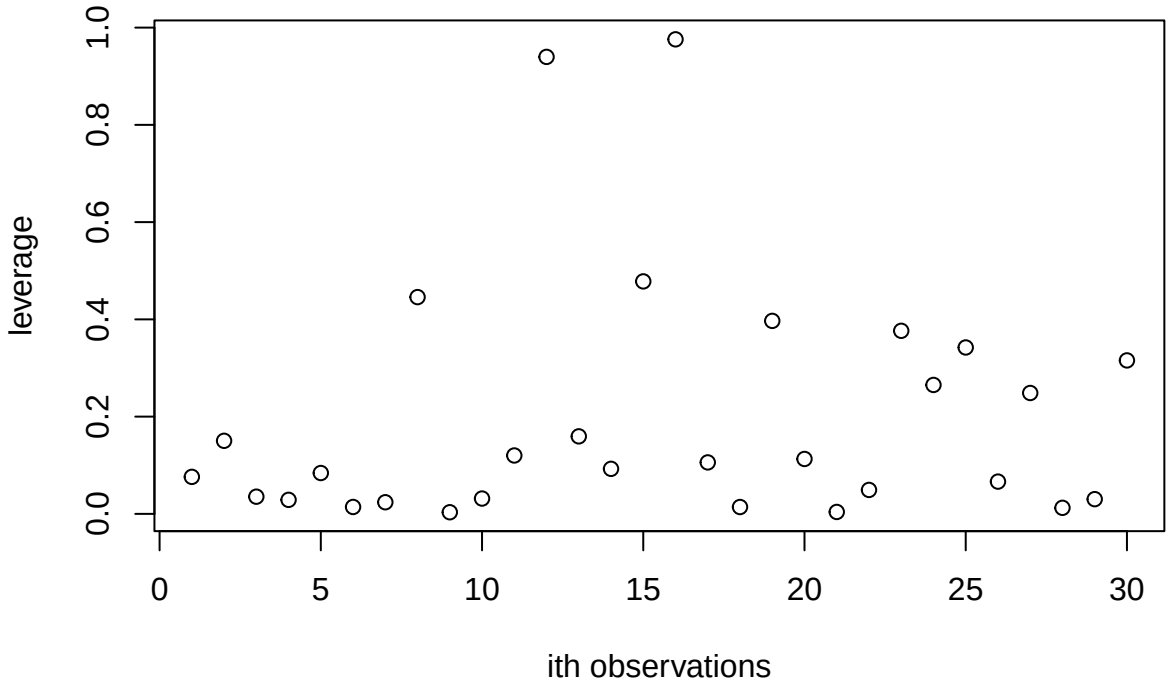
```
##
## Call:
## glm(formula = ymatrix ~ ., family = famil, data = data.frame(ymatrix,
##     xmatrix))
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -15.337942   9.423550  -1.628    0.117
## Endemics      4.393654   0.481203   9.131 4.13e-09 ***
## Area          0.013258   0.011403   1.163    0.257
## Elevation    -0.047537   0.047596  -0.999    0.328
## Nearest      -0.101460   0.500871  -0.203    0.841
## Scruz         0.008256   0.105884   0.078    0.939
## Adjacent      0.001811   0.011879   0.152    0.880
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 838.9007)
##
##     Null deviance: 381081  on 29  degrees of freedom
## Residual deviance:  19295  on 23  degrees of freedom
## AIC: 295.13
##
## Number of Fisher Scoring iterations: 2
```


Fitted linear predictor
0
100
200
300
−3
−1
0
1
2
3
Standardised deviance


Fitted mean
0
100
200
300
−3
−1
0
1
2
3
Standardised deviance


covariate 1
0
80
−1400
−1000
−600
−200
0
ei
covariate 2
0
0
5000
10000
15000
20000
ei
covariate 3
0
−60
−40
−20
0
20
40
60
80
ei
covariate 4
0
40
−60
−40
−20
0
20
40
60
ei
covariate 5
0
250
−60
−40
−20
0
20
40
60
ei
covariate 6
0
−60
−40
−20
0
20
40
60
ei


cook's distance
0
10
20
30
40
50
0
5
10
15
20
25
30
ith observations



Remarks:

1. Both plots should show a similar trend because the fitted linear predictor η and the fitted mean μ are related by the smooth link function $\eta = f(\mu)$. However, they are plotted on different scales, so they do not have to look identical.
2. If the model is selected correctly, the residuals should be centered around 0 with roughly constant variance. We should not expect any curvature or systematic pattern.
3. For simplicity, we assume the GLM is either binomial, gamma, normal, or Poisson, with the corresponding link function.

Remark 3 continue: For a Poisson GLM with the log link,

$$g(\mu_i) = \eta_i = \log(\mu_i)$$

Therefore, the adjusted dependent variable is

$$z_i = \hat{\eta}_i + (y_i - \hat{\mu}_i)\left(\frac{1}{\hat{\mu}_i}\right) = \log(\hat{\mu}_i) + \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i}$$

For a Gamma GLM with the inverse link,

$$g(\mu_i) = \eta_i = \frac{1}{\mu_i}$$

Similarly,

$$z_i = \frac{1}{\hat{\mu}_i} - \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i^2}.$$

For a Normal GLM with the identity link,

$$g(\mu_i) = \eta_i = \mu_i$$

Similarly,

$$z_i = \hat{\mu}_i + (y_i - \hat{\mu}_i) = y_i.$$

For a Binomial GLM with the logit link,

$$g(\mu_i) = \eta_i = \log\left(\frac{\mu_i}{1-\mu_i}\right)$$

Similarly,

$$z_i = \log\left(\frac{\hat{\mu}_i}{1-\hat{\mu}_i}\right) + \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i(1-\hat{\mu}_i)}.$$

4. If the scale is correct, the observations should lie close to the line of best fit. If the plot shows clear curvature rather than an approximately linear relationship, then the link scale may not be suitable for the observations.
5. Cook's distance measures how much influence an observation has on the model. If Cook's distance is unexpectedly large, the observation may have too much influence on the fitted model, so caution is needed when interpreting the results.
6. Leverage indicates how much an observation can potentially influence the fitted GLM because of its unusual predictor values.